



810 Invoice

X12/V4010/810 : 810 Invoice

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Notes

EDI notes

The EDI messages use UTF8/Unicode characters and the field data is limited to the following characters:

A through Z (upper or lowercase) (non-English characters are not permitted)

0 through 9

! " & ' () / + , - . : ; = (space) % @ [] _ { } # \$

All messages between trading partners need to meet EDI validation.

Upon receiving an EDI message D&H will validate the message and return a 997 Functional Acknowledgment.

The sample segments and messages included in this guide use the asterisk as the data element separator. The separators used for actual EDI transmissions will be agreed upon with each trading partner.

Note- D&H requires any segment reflected in this guide as 'mandatory' (note under the usage section) to be transmitted in your documents. We will accept all valid X12 segments (By ANSI standards) but will only process those required for our business unless otherwise negotiated during your implementation

EDI Contacts

General EDI Email: edi@dandh.com

For additional processing information, please refer to the following documents:

US Terms and Conditions (<https://www.dandh.com/docs/vendor/termsandconditions.pdf>)

CA Terms and Conditions (<https://www.dandh.ca/docs/vendor/termsandconditions.pdf>)

US Routing Guide: (<https://www.dandh.com/docs/vendor/US-DomesticSupplierRoutingGuide.pdf>)

CA Routing Guide: (<https://www.dandh.ca/docs/vendor/CA-DomesticSupplierRoutingGuide.pdf>)

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810 Invoice

Functional Group=IN

Purpose: This Draft Standard for Trial Use contains the format and establishes the data contents of the Invoice Transaction Set (810) for use within the context of an Electronic Data Interchange (EDI) environment. The transaction set can be used to provide for customary and established business and industry practice relative to the billing for goods and services provided.

Not Defined:

<u>Pos</u>	<u>Id</u>	<u>Segment Name</u>	<u>Req</u>	<u>Max Use</u>	<u>Repeat</u>	<u>Notes</u>	<u>Usage</u>
	ISA	Interchange Control Header	M	1			Must use
	GS	Functional Group Header	M	1			Must use

Heading:

<u>Pos</u>	<u>Id</u>	<u>Segment Name</u>	<u>Req</u>	<u>Max Use</u>	<u>Repeat</u>	<u>Notes</u>	<u>Usage</u>
010	ST	Transaction Set Header	M	1			Must use
020	BIG	Beginning Segment for Invoice	M	1			Must use
040	CUR	Currency	O	1			Must use
050	REF	Reference Identification	O	12			Must use
LOOP ID - N1					200		
070	N1	Name	O	1			Must use
090	N3	Address Information	O	2			Must use
100	N4	Geographic Location	O	1			Must use
130	ITD	Terms of Sale/Deferred Terms of Sale	O	>1			Must use

Detail:

<u>Pos</u>	<u>Id</u>	<u>Segment Name</u>	<u>Req</u>	<u>Max Use</u>	<u>Repeat</u>	<u>Notes</u>	<u>Usage</u>
LOOP ID - IT1					200000		
010	IT1	Baseline Item Data (Invoice)	O	1			Must use

Summary:

<u>Pos</u>	<u>Id</u>	<u>Segment Name</u>	<u>Req</u>	<u>Max Use</u>	<u>Repeat</u>	<u>Notes</u>	<u>Usage</u>
010	TDS	Total Monetary Value Summary	M	1			Must use
020	TXI	Tax Information	O	10			Must use
LOOP ID - SAC					25		
040	SAC	Service, Promotion, Allowance, or Charge Information	O	1			Must use
070	CTT	Transaction Totals	O	1		N3/070	Must use
080	SE	Transaction Set Trailer	M	1			Must use
	GE	Functional Group Trailer	M	1			Must use
	IEA	Interchange Control Trailer	M	1			Must use

Notes:

3/070 Number of line items (CTT01) is the accumulation of the number of IT1 segments. If used, hash total (CTT02) is the sum of the value of quantities invoiced (IT102) for each IT1 segment.

D & H Notes 1:

Summary of Changes

03/01/2015 Initial Version

02/26/2016 Added N1 Loops

ISA Interchange Control Header

Pos:	Max: 1
Not Defined - Mandatory	
Loop: N/A	Elements: 16

User Option (Usage): Must use

Purpose: To start and identify an interchange of zero or more functional groups and interchange-related control segments

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
ISA01	I01	Authorization Information Qualifier	M	ID	2/2	Must use
		Description: Code to identify the type of information in the Authorization Information				
		<u>Code</u> <u>Name</u> 00 No Authorization Information Present (No Meaningful Information in I02)				
ISA02	I02	Authorization Information	M	AN	10/10	Must use
		Description: Information used for additional identification or authorization of the interchange sender or the data in the interchange; the type of information is set by the Authorization Information Qualifier (I01)				
ISA03	I03	Security Information Qualifier	M	ID	2/2	Must use
		Description: Code to identify the type of information in the Security Information				
		<u>Code</u> <u>Name</u> 00 No Security Information Present (No Meaningful Information in I04)				
ISA04	I04	Security Information	M	AN	10/10	Must use
		Description: This is used for identifying the security information about the interchange sender or the data in the interchange; the type of information is set by the Security Information Qualifier (I03)				
ISA05	I05	Interchange ID Qualifier	M	ID	2/2	Must use
		Description: Qualifier to designate the system/method of code structure used to designate the sender or receiver ID element being qualified				
		<u>Code</u> <u>Name</u> 01 Duns (Dun & Bradstreet) 08 UCC EDI Communications ID (Comm ID) 12 Phone (Telephone Companies) ZZ Mutually Defined				
ISA06	I06	Interchange Sender ID	M	AN	15/15	Must use

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		Description: Identification code published by the sender for other parties to use as the receiver ID to route data to them; the sender always codes this value in the sender ID element D & H Use/Samples: 007911209-D&H Production ID 007911209T-D & H Test ID 2433				
ISA07	I05	Interchange ID Qualifier Description: Qualifier to designate the system/method of code structure used to designate the sender or receiver ID element being qualified D & H Use/Samples: 01-D&H US 08-D & H Canada <u>Code</u> <u>Name</u> 01 Duns (Dun & Bradstreet) 08 UCC EDI Communications ID (Comm ID)	M	ID	2/2	Must use
ISA08	I07	Interchange Receiver ID Description: Identification code published by the receiver of the data; When sending, it is used by the sender as their sending ID, thus other parties sending to them will use this as a receiving ID to route data to them D & H Use/Samples: 007911209-US Production ID 007911209T-US Production ID 243337024 -Canada Production ID 243337024T-Canada Test ID	M	AN	15/15	Must use
ISA09	I08	Interchange Date Description: Date of the interchange	M	DT	6/6	Must use
ISA10	I09	Interchange Time Description: Time of the interchange	M	TM	4/4	Must use
ISA11	I10	Interchange Control Standards Identifier Description: Code to identify the agency responsible for the control standard used	M	ID	1/1	Must use

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		by the message that is enclosed by the interchange header and trailer				
		<u>Code</u> <u>Name</u>				
		U U.S. EDI Community of ASC X12, TDCC, and UCS				
ISA12	I11	Interchange Control Version Number	M	ID	5/5	Must use
		Description: Code specifying the version number of the interchange control segments				
		<u>Code</u> <u>Name</u>				
		00401 Draft Standards for Trial Use Approved for Publication by ASC X12 Procedures Review Board through October 1997				
ISA13	I12	Interchange Control Number	M	N0	9/9	Must use
		Description: A control number assigned by the interchange sender				
ISA14	I13	Acknowledgment Requested	M	ID	1/1	Must use
		Description: Code sent by the sender to request an interchange acknowledgment (TA1)				
		<u>Code</u> <u>Name</u>				
		0 No Acknowledgment Requested				
ISA15	I14	Usage Indicator	M	ID	1/1	Must use
		Description: Code to indicate whether data enclosed by this interchange envelope is test, production or information				
		<u>Code</u> <u>Name</u>				
		P Production Data				
ISA16	I15	Component Element Separator	M		1/1	Must use
		Description: Type is not applicable; the component element separator is a delimiter and not a data element; this field provides the delimiter used to separate component data elements within a composite data structure; this value must be different than the data element separator and the segment terminator				

D & H Use/Samples:

```
ISA*00*      *00*      *ZZ*BROTHERUSA  *01*007911209  *160301*0556*U*00401*000002196*0*P*>
```


GS Functional Group Header

Pos:	Max: 1
Not Defined - Mandatory	
Loop: N/A	Elements: 8

User Option (Usage): Must use

Purpose: To indicate the beginning of a functional group and to provide control information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
GS01	479	Functional Identifier Code	M	ID	2/2	Must use
		Description: Code identifying a group of application related transaction sets				
		Code Name IN Invoice Information (810,819)				
GS02	142	Application Sender's Code	M	AN	2/15	Must use
		Description: Code identifying party sending transmission; codes agreed to by trading partners				
		D & H Use/Samples: <i>007911209-US Production Id</i> <i>007911209T-US Test ID</i> <i>243337024-Canada Production ID</i> <i>243337024T-Canada Test ID</i>				
GS03	124	Application Receiver's Code	M	AN	2/15	Must use
		Description: Code identifying party receiving transmission; codes agreed to by trading partners				
GS04	373	Date	M	DT	8/8	Must use
		Description: Date expressed as CCYYMMDD				
GS05	337	Time	M	TM	4/8	Must use
		Description: Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99)				
GS06	28	Group Control Number	M	N0	1/9	Must use
		Description: Assigned number originated and maintained by the sender				
GS07	455	Responsible Agency Code	M	ID	1/2	Must use

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		Description: Code identifying the issuer of the standard; this code is used in conjunction with Data Element 480				
		Code Name X Accredited Standards Committee X12				
GS08	480	Version / Release / Industry Identifier Code	M	AN	1/12	Must use
		Description: Code indicating the version, release, subrelease, and industry identifier of the EDI standard being used, including the GS and GE segments; if code in DE455 in GS segment is X, then in DE 480 positions 1-3 are the version number; positions 4-6 are the release and subrelease, level of the version; and positions 7-12 are the industry or trade association identifiers (optionally assigned by user); if code in DE455 in GS segment is T, then other formats are allowed				
		Code Name 004010 Draft Standards Approved for Publication by ASC X12 Procedures Review Board through October 1997				

Semantics:

1. GS04 is the group date.
2. GS05 is the group time.
3. The data interchange control number GS06 in this header must be identical to the same data element in the associated functional group trailer, GE02.

Comments:

1. A functional group of related transaction sets, within the scope of X12 standards, consists of a collection of similar transaction sets enclosed by a functional group header and a functional group trailer.

D & H Use/Samples:

GS*IN*BROTHERUSA*007911209*20160301*0556*2196*X*004010

ST Transaction Set Header

Pos: 010	Max: 1
Heading - Mandatory	
Loop: N/A	Elements: 2

User Option (Usage): Must use

Purpose: To indicate the start of a transaction set and to assign a control number

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
ST01	143	Transaction Set Identifier Code	M	ID	3/3	Must use
Description: Code uniquely identifying a Transaction Set						
<u>Code</u> <u>Name</u> 810 Invoice						
ST02	329	Transaction Set Control Number	M	AN	4/9	Must use
Description: Identifying control number that must be unique within the transaction set functional group assigned by the originator for a transaction set						

Semantics:

1. The transaction set identifier (ST01) used by the translation routines of the interchange partners to select the appropriate transaction set definition (e.g., 810 selects the Invoice Transaction Set).

D & H Use/Samples:

ST*810*0001

BIG Beginning Segment for Invoice

Pos: 020	Max: 1
Heading - Mandatory	
Loop: N/A	Elements: 4

User Option (Usage): Must use

Purpose: To indicate the beginning of an invoice transaction set and transmit identifying numbers and dates

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
BIG01	373	Date	M	DT	8/8	Must use
		Description: Date expressed as CCYYMMDD				
BIG02	76	Invoice Number	M	AN	1/22	Must use
		Description: Identifying number assigned by issuer				
BIG03	373	Date	O	DT	8/8	Must use
		Description: Date expressed as CCYYMMDD				
BIG04	324	Purchase Order Number	O	AN	1/22	Must use
		Description: Identifying number for Purchase Order assigned by the orderer/purchaser				

Semantics:

1. BIG01 is the invoice issue date.
2. BIG03 is the date assigned by the purchaser to purchase order.
3. BIG10 indicates the consolidated invoice number. When BIG07 contains code CI, BIG10 is not used.

Comments:

1. BIG07 is used only to further define the type of invoice when needed.

D & H Use/Samples:

*BIG*20160229*0212132359*20160223*917871L*

CUR Currency

Pos: 040	Max: 1
Heading - Optional	
Loop: N/A	Elements: 2

User Option (Usage): Must use

Purpose: To specify the currency (dollars, pounds, francs, etc.) used in a transaction

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
CUR01	98	Entity Identifier Code	M	ID	2/3	Must use
Description: Code identifying an organizational entity, a physical location, property or an individual						
Code Name						
BY Buying Party (Purchaser)						
CUR02	100	Currency Code	M	ID	3/3	Must use
Description: Code (Standard ISO) for country in whose currency the charges are specified						

Syntax Rules:

1. C0807 - If CUR08 is present, then CUR07 is required.
2. C0907 - If CUR09 is present, then CUR07 is required.
3. L101112 - If CUR10 is present, then at least one of CUR11 or CUR12 is required.
4. C1110 - If CUR11 is present, then CUR10 is required.
5. C1210 - If CUR12 is present, then CUR10 is required.
6. L131415 - If CUR13 is present, then at least one of CUR14 or CUR15 is required.
7. C1413 - If CUR14 is present, then CUR13 is required.
8. C1513 - If CUR15 is present, then CUR13 is required.
9. L161718 - If CUR16 is present, then at least one of CUR17 or CUR18 is required.
10. C1716 - If CUR17 is present, then CUR16 is required.
11. C1816 - If CUR18 is present, then CUR16 is required.
12. L192021 - If CUR19 is present, then at least one of CUR20 or CUR21 is required.
13. C2019 - If CUR20 is present, then CUR19 is required.
14. C2119 - If CUR21 is present, then CUR19 is required.

Comments:

1. See Figures Appendix for examples detailing the use of the CUR segment.

D & H Use/Samples:

CUR*BY*USD

REF Reference Identification

Pos: 050	Max: 12
Heading - Optional	
Loop: N/A	Elements: 2

User Option (Usage): Must use

Purpose: To specify identifying information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
REF01	128	Reference Identification Qualifier	M	ID	2/3	Must use
		Description: Code qualifying the Reference Identification				
		<u>Code</u> <u>Name</u>				
		IA Internal Vendor Number				
REF02	127	Reference Identification	X	AN	1/30	Must use
		Description: Reference information as defined for a particular Transaction Set or as specified by the Reference Identification Qualifier				

Syntax Rules:

1. R0203 - At least one of REF02 or REF03 is required.

Semantics:

1. REF04 contains data relating to the value cited in REF02.

D & H Use/Samples:

N3*3701 S MINNEWAWA AVENUE

N1 Name

Pos: 070	Max: 1
Heading - Optional	
Loop: N1	Elements: 4

User Option (Usage): Must use

Purpose: To identify a party by type of organization, name, and code

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N101	98	Entity Identifier Code	M	ID	2/3	Must use
		Description: Code identifying an organizational entity, a physical location, property or an individual				
		Code Name BT Bill-to-Party EN End User RI Remit To ST Ship To				
N102	93	Name	X	AN	1/60	Must use
		Description: Free-form name				
N103	66	Identification Code Qualifier	X	ID	1/2	Used
		Description: Code designating the system/method of code structure used for Identification Code (67)				
		Code Name 91 Assigned by Seller or Seller's Agent 92 Assigned by Buyer or Buyer's Agent				
N104	67	Identification Code	X	AN	2/80	Used
		Description: Code identifying a party or other code				

Syntax Rules:

1. R0203 - At least one of N102 or N103 is required.
2. P0304 - If either N103 or N104 is present, then the other is required.

Comments:

1. This segment, used alone, provides the most efficient method of providing organizational identification. To obtain this efficiency the "ID Code" (N104) must provide a key to the table maintained by the transaction processing party.
2. N105 and N106 further define the type of entity in N101.

N3 Address Information

Pos: 090	Max: 2
Heading - Optional	
Loop: N1	Elements: 2

User Option (Usage): Must use

Purpose: To specify the location of the named party

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N301	166	Address Information	M	AN	1/55	Must use
		Description: Address information				
N302	166	Address Information	O	AN	1/55	Used
		Description: Address information				

D & H Use/Samples:

N3*3701 S MINNEWAWA AVENUE

N4 Geographic Location

Pos: 100	Max: 1
Heading - Optional	
Loop: N1	Elements: 4

User Option (Usage): Must use

Purpose: To specify the geographic place of the named party

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
N401	19	City Name	O	AN	2/30	Must use
		Description: Free-form text for city name				
N402	156	State or Province Code	O	ID	2/2	Must use
		Description: Code (Standard State/Province) as defined by appropriate government agency				
N403	116	Postal Code	O	ID	3/15	Must use
		Description: Code defining international postal zone code excluding punctuation and blanks (zip code for United States)				
N404	26	Country Code	O	ID	2/3	Must use
		Description: Code identifying the country				

Syntax Rules:

1. C0605 - If N406 is present, then N405 is required.

Comments:

1. A combination of either N401 through N404, or N405 and N406 may be adequate to specify a location.
2. N402 is required only if city name (N401) is in the U.S. or Canada.

ITD Terms of Sale/Deferred Terms of Sale

Pos: 130	Max: >1
Heading - Optional	
Loop: N/A	Elements: 11

User Option (Usage): Must use

Purpose: To specify terms of sale

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
ITD01	336	Terms Type Code	O	ID	2/2	Must use
		Description: Code identifying type of payment terms				
		Code Name				
		01 Basic				
ITD02	333	Terms Basis Date Code	O	ID	1/2	Must use
		Description: Code identifying the beginning of the terms period				
		Code Name				
		1 Ship Date				
		3 Invoice Date				
ITD03	338	Terms Discount Percent	O	R	1/6	Used
		Description: Terms discount percentage, expressed as a percent, available to the purchaser if an invoice is paid on or before the Terms Discount Due Date				
ITD05	351	Terms Discount Days Due	X	N0	1/3	Must use
		Description: Number of days in the terms discount period by which payment is due if terms discount is earned				
ITD06	446	Terms Net Due Date	O	DT	8/8	Must use
		Description: Date when total invoice amount becomes due expressed in format CCYYMMDD				
ITD07	386	Terms Net Days	O	N0	1/3	Must use
		Description: Number of days until total invoice amount is due (discount not applicable)				
ITD08	362	Terms Discount Amount	O	N2	1/10	Used
		Description: Total amount of terms discount				
ITD09	388	Terms Deferred Due Date	O	DT	8/8	Used
		Description: Date deferred payment or percent of invoice payable is due				

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
		expressed in format CCYYMMDD				
ITD12	352	Description	O	AN	1/80	Must use
		Description: A free-form description to clarify the related data elements and their content				
ITD15	954	Percent	O	R	1/10	Used
		Description: Percentage expressed as a decimal				

Syntax Rules:

1. L03040513 - If ITD03 is present, then at least one of ITD04, ITD05 or ITD13 is required.
2. L08040513 - If ITD08 is present, then at least one of ITD04, ITD05 or ITD13 is required.
3. L091011 - If ITD09 is present, then at least one of ITD10 or ITD11 is required.

Semantics:

1. ITD15 is the percentage applied to a base amount used to determine a late payment charge.

Comments:

1. If the code in ITD01 is "04", then ITD07 or ITD09 is required and either ITD10 or ITD11 is required; if the code in ITD01 is "05", then ITD06 or ITD07 is required.

D & H Use/Samples:

*ITD*01*3*2*20160331*30*20160429*60*1515*20160429***Up to 04/29/2016 without deduction*

IT1 Baseline Item Data (Invoice)

Pos: 010	Max: 1
Detail - Optional	
Loop: IT1	Elements: 10

User Option (Usage): Must use

Purpose: To specify the basic and most frequently used line item data for the invoice and related transactions

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
IT101	350	Assigned Identification Description: Alphanumeric characters assigned for differentiation within a transaction set	O	AN	1/20	Must use
IT102	358	Quantity Invoiced Description: Number of units invoiced (supplier units)	X	R	1/10	Used
IT103	355	Unit or Basis for Measurement Code Description: Code specifying the units in which a value is being expressed, or manner in which a measurement has been taken <u>Code</u> <u>Name</u> EA Each	X	ID	2/2	Must use
IT104	212	Unit Price Description: Price per unit of product, service, commodity, etc.	X	R	1/17	Must use
IT106	235	Product/Service ID Qualifier Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234) <u>Code</u> <u>Name</u> VP Vendor's (Seller's) Part Number	X	ID	2/2	Must use
IT107	234	Product/Service ID Description: Identifying number for a product or service	X	AN	1/48	Must use
IT108	235	Product/Service ID Qualifier Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234) <u>Code</u> <u>Name</u> BP Buyer's Part Number	X	ID	2/2	Must use
IT109	234	Product/Service ID Description: Identifying number for a	X	AN	1/48	Must use

<u>Ref</u>	<u>Id</u>	<u>Element Name</u> product or service	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
IT110	235	Product/Service ID Qualifier Description: Code identifying the type/source of the descriptive number used in Product/Service ID (234)	X	ID	2/2	Used
		Code Name UP U.P.C. Consumer Package Code (1-5-5-1)				
IT111	234	Product/Service ID Description: Identifying number for a product or service	X	AN	1/48	Used

Syntax Rules:

1. P020304 - If either IT102, IT103 or IT104 are present, then the others are required.
2. P0607 - If either IT106 or IT107 is present, then the other is required.
3. P0809 - If either IT108 or IT109 is present, then the other is required.
4. P1011 - If either IT110 or IT111 is present, then the other is required.
5. P1213 - If either IT112 or IT113 is present, then the other is required.
6. P1415 - If either IT114 or IT115 is present, then the other is required.
7. P1617 - If either IT116 or IT117 is present, then the other is required.
8. P1819 - If either IT118 or IT119 is present, then the other is required.
9. P2021 - If either IT120 or IT121 is present, then the other is required.
10. P2223 - If either IT122 or IT123 is present, then the other is required.
11. P2425 - If either IT124 or IT125 is present, then the other is required.

Semantics:

1. IT101 is the purchase order line item identification.

Comments:

1. Element 235/234 combinations should be interpreted to include products and/or services. See the Data Dictionary for a complete list of IDs.
2. IT106 through IT125 provide for ten different product/service IDs for each item. For example: Case, Color, Drawing No., U.P.C. No., ISBN No., Model No., or SKU.

D & H Use/Samples:

IT1*000009*12*EA*14.24**VP*DK1209*BP*DK1209*UP*012502611653

D & H Use/Samples 2:

Implementation Notes Please use correct qualifiers for D&H Item and Manufacturers item as shown above. UPC is optional.

The D&H and Manufacturers items must be identical to the items reported on the purchase order being invoiced.

TDS Total Monetary Value Summary

Pos: 010	Max: 1
Summary - Mandatory	
Loop: N/A	Elements: 1

User Option (Usage): Must use

Purpose: To specify the total invoice discounts and amounts

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
TDS01	610	Amount	M	N2	1/15	Must use

Description: Monetary amount

Semantics:

1. TDS01 is the total amount of invoice (including charges, less allowances) before terms discount (if discount is applicable).
2. TDS02 indicates the amount upon which the terms discount amount is calculated.
3. TDS03 is the amount of invoice due if paid by terms discount due date (total invoice or installment amount less cash discount).
4. TDS04 indicates the total amount of terms discount.

Comments:

1. TDS02 is required if the dollar value subject to discount is not equal to the dollar value of TDS01.

D & H Use/Samples:

TDS*24363068

TXI Tax Information

Pos: 020	Max: 10
Summary - Optional	
Loop: N/A	Elements: 3

User Option (Usage): Must use

Purpose: To specify tax information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
TXI01	963	Tax Type Code	M	ID	2/2	Must use
		Description: Code specifying the type of tax				
		<u>Code</u> <u>Name</u>				
		GS Goods and Services Tax				
		Description: <i>Canadian value-added tax</i>				
		PG State or Provincial Tax on Goods				
TXI02	782	Monetary Amount	X	R	1/18	Must use
		Description: Monetary amount				
TXI03	954	Percent	X	R	1/10	Must use
		Description: Percentage expressed as a decimal				

Syntax Rules:

1. R020306 - At least one of TXI02, TXI03 or TXI06 is required.
2. P0405 - If either TXI04 or TXI05 is present, then the other is required.
3. C0803 - If TXI08 is present, then TXI03 is required.

Semantics:

1. TXI02 is the monetary amount of the tax.
2. TXI03 is the tax percent expressed as a decimal.
3. TXI07 is a code indicating the relationship of the price or amount to the associated segment.

D & H Use/Samples:

TXI*GS*74.61*6

SAC Service, Promotion, Allowance, or Charge Information

Pos: 040	Max: 1
Summary - Optional	
Loop: SAC	Elements: 7

User Option (Usage): Must use

Purpose: To request or identify a service, promotion, allowance, or charge; to specify the amount or percentage for the service, promotion, allowance, or charge

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
SAC01	248	Allowance or Charge Indicator	M	ID	1/1	Must use
		Description: Code which indicates an allowance or charge for the service specified				
		Code Name				
		A Allowance				
		C Charge				
SAC02	1300	Service, Promotion, Allowance, or Charge Code	X	ID	4/4	Must use
		Description: Code identifying the service, promotion, allowance, or charge				
		Code Name				
		C350 Distributor Discount/Allowance				
		D240 Freight				
SAC05	610	Amount	O	N2	1/15	Must use
		Description: Monetary amount				
SAC09	355	Unit or Basis for Measurement Code	X	ID	2/2	Used
		Description: Code specifying the units in which a value is being expressed, or manner in which a measurement has been taken				
SAC10	380	Quantity	X	R	1/15	Used
		Description: Numeric value of quantity				
SAC12	331	Allowance or Charge Method of Handling Code	O	ID	2/2	Must use
		Description: Code indicating method of handling for an allowance or charge				
		Code Name				
		01 Bill Back				
		02 Off Invoice				
		03 Vendor Check to Customer				
		04 Credit Customer Account				

		<u>Code</u>	<u>Name</u>				
		05	Charge to be Paid by Vendor				
		06	Charge to be Paid by Customer				
SAC15	352	Description		X	AN	1/80	Must use
Description: A free-form description to clarify the related data elements and their content							

Syntax Rules:

1. R0203 - At least one of SAC02 or SAC03 is required.
2. P0304 - If either SAC03 or SAC04 is present, then the other is required.
3. P0607 - If either SAC06 or SAC07 is present, then the other is required.
4. P0910 - If either SAC09 or SAC10 is present, then the other is required.
5. C1110 - If SAC11 is present, then SAC10 is required.
6. L130204 - If SAC13 is present, then at least one of SAC02 or SAC04 is required.
7. C1413 - If SAC14 is present, then SAC13 is required.
8. C1615 - If SAC16 is present, then SAC15 is required.

Semantics:

1. If SAC01 is "A" or "C", then at least one of SAC05, SAC07, or SAC08 is required.
2. SAC05 is the total amount for the service, promotion, allowance, or charge.
3. If SAC05 is present with SAC07 or SAC08, then SAC05 takes precedence.
4. SAC08 is the allowance or charge rate per unit.
5. SAC10 and SAC11 is the quantity basis when the allowance or charge quantity is different from the purchase order or invoice quantity.
6. SAC10 and SAC11 used together indicate a quantity range, which could be a dollar amount, that is applicable to service, promotion, allowance, or charge.
7. SAC13 is used in conjunction with SAC02 or SAC04 to provide a specific reference number as identified by the code used.
8. SAC14 is used in conjunction with SAC13 to identify an option when there is more than one option of the promotion.
9. SAC16 is used to identify the language being used in SAC15.

Comments:

1. SAC04 may be used to uniquely identify the service, promotion, allowance, or charge. In addition, it may be used in conjunction to further the code in SAC02.
2. In some business applications, it is necessary to advise the trading partner of the actual dollar amount that a particular allowance, charge, or promotion was based on to reduce ambiguity. This amount is commonly referred to as "Dollar Basis Amount". It is represented in the SAC segment in SAC10 using the qualifier "DO" - Dollars in SAC09.

CTT Transaction Totals

Pos: 070	Max: 1
Summary - Optional	
Loop: N/A	Elements: 1

User Option (Usage): Must use

Purpose: To transmit a hash total for a specific element in the transaction set

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
CTT01	354	Number of Line Items	M	N0	1/6	Must use

Description: Total number of line items in the transaction set

Syntax Rules:

1. P0304 - If either CTT03 or CTT04 is present, then the other is required.
2. P0506 - If either CTT05 or CTT06 is present, then the other is required.

Comments:

1. This segment is intended to provide hash totals to validate transaction completeness and correctness.

D & H Use/Samples:

CTT*101

SE Transaction Set Trailer

Pos: 080	Max: 1
Summary - Mandatory	
Loop: N/A	Elements: 2

User Option (Usage): Must use

Purpose: To indicate the end of the transaction set and provide the count of the transmitted segments (including the beginning (ST) and ending (SE) segments)

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
SE01	96	Number of Included Segments	M	N0	1/10	Must use
		Description: Total number of segments included in a transaction set including ST and SE segments				
SE02	329	Transaction Set Control Number	M	AN	4/9	Must use
		Description: Identifying control number that must be unique within the transaction set functional group assigned by the originator for a transaction set				

Comments:

1. SE is the last segment of each transaction set.

GE Functional Group Trailer

Pos:	Max: 1
Summary - Mandatory	
Loop: N/A	Elements: 2

User Option (Usage): Must use

Purpose: To indicate the end of a functional group and to provide control information

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
GE01	97	Number of Transaction Sets Included	M	N0	1/6	Must use
		Description: Total number of transaction sets included in the functional group or interchange (transmission) group terminated by the trailer containing this data element				
GE02	28	Group Control Number	M	N0	1/9	Must use
		Description: Assigned number originated and maintained by the sender				

Semantics:

1. The data interchange control number GE02 in this trailer must be identical to the same data element in the associated functional group header, GS06.

Comments:

1. The use of identical data interchange control numbers in the associated functional group header and trailer is designed to maximize functional group integrity. The control number is the same as that used in the corresponding header.

D & H Use/Samples:

GE*6*2196

IEA Interchange Control Trailer

Pos:	Max: 1
Summary - Mandatory	
Loop: N/A	Elements: 2

User Option (Usage): Must use

Purpose: To define the end of an interchange of zero or more functional groups and interchange-related control segments

Element Summary:

<u>Ref</u>	<u>Id</u>	<u>Element Name</u>	<u>Req</u>	<u>Type</u>	<u>Min/Max</u>	<u>Usage</u>
IEA01	I16	Number of Included Functional Groups	M	N0	1/5	Must use
		Description: A count of the number of functional groups included in an interchange				
IEA02	I12	Interchange Control Number	M	N0	9/9	Must use
		Description: A control number assigned by the interchange sender				

D & H Use/Samples:

```
ISA*00*      *00*      *ZZ*BROTHERUSA  *01*007911209  *160301*0556*U*00401*000002196*0*P*>~
GS*IN*BROTHERUSA*007911209*20160301*0556*2196*X*004010~
ST*810*0001~
BIG*20160229*0212132359*20160223*917871L~
CUR*BY*USD~
REF*IA*0004948081~
N1*RI*BROTHER INT CORP~
N3*100 SOMERSET CORP~
N4*BRIDGEWATER*NJ*08807*US~
N1*BT*D & H DISTRIBUTING CO.*91*141889~
N3*2525 N SEVENTH ST~
N4*HARRISBURG*PA*171102511*US~
N1*ST*D&H DISTRIBUTING COMPANY*92*04~
N3*3701 S MINNEWAWA AVENUE~
N4*FRESNO*CA*93725*US~
N1*EN*D&H DISTRIBUTING COMPANY*92*04~
N3*3701 S MINNEWAWA AVENUE~
N4*FRESNO*CA*93725*US~
ITD*01*3*2*20160331*30*20160429*60*1515*20160429***Up to 04/29/2016 without deduction**350~
IT1*000001*12*EA*123.59**VP*DR400*BP*DR400*UP*012502525967~
IT1*000002*8*EA*415**VP*PPF4750E*BP*PPF4750E*UP*012502602835~
IT1*000003*24*EA*310**VP*PT9600*BP*PT9600*UP*012502602774~
IT1*000004*24*EA*7.72**VP*DK1201*BP*DK1201*UP*012502611639~
IT1*000005*24*EA*11.87**VP*DK1218*BP*DK1218*UP*012502613527~
IT1*000006*64*EA*5.65**VP*M231*BP*M231*UP*012502053736~
IT1*000007*12*EA*96.86**VP*DR620*BP*DR620RDU*UP*012502622291~
IT1*000008*24*EA*45.21**VP*TN210C*BP*TN210C*UP*012502622598~
IT1*000009*12*EA*14.24**VP*DK1209*BP*DK1209*UP*012502611653~
```

TDS*24363068~
 TXI*GS*74.61*6~
 SAC*A*C350***100*****02***COST OF GOODS ALLOWANCE~
 SAC*C*D240***5742*****02***FREIGHT CHARGE~
 CTT*101~
 SE*223*0006~
 GE*6*2196~
 IEA*1*000002196~

D & H Use/Samples 2:

IEA*1*000002196

Appendix B

D&H Distribution Facilities

The following is a list of the distribution facilities for D&H and the expected EDI message format for the addresses of the facilities

Katie Court – Branch 1 909 Katie Court Harrisburg, PA 17109 N1*ST*D&H - HARRISBURG*92*01 N3*UNION SQUARE IND PK*909 KATIE COURT N4*Harrisburg*PA*17109*US	Fresno – Branch 4 3701 South Minnewawa Avenue Fresno, CA 93725 N1*ST*D&H - FRESNO*92*04 N3*3701 S. Minnewawa Ave N4*Fresno*CA*93725*US
Camp Hill (Returns) – Branch 2 500 Terminal Road Camp Hill, PA 17011 N1*ST*D&H – CAMP HILL*92*02 N3*500 Terminal Road N4*Camp Hill*PA*17011*US	Chicago – Branch 5 1455A Remington Boulevard Bolingbrook, IL 90490 N1*ST*D&H - CHICAGO*92*05 N3*1455A Remington Boulevard N4*Bolingbrook*IL*60490*US
Canada – Branch 3 7975 Heritage Road, Unit 20 Brampton, Ontario L6Y5X5 ST*D&H - CANADA*92*03 N3*7975 Heritage Road*Unit 20 Building A N4*Brampton*ON*L6Y 5X5*CA	Atlanta/Newnan – Branch 6 185 Coweta Industrial Parkway Newnan, GA 30265 N1*ST*D&H - ATLANTA*92*06 N3*185 Coweta Industrial Parkway N4*Newnan*GA*30265*US